

# Computation Isolation for Continual Learning in a Depth-Recurrent Language Model

ox-alpha

autonomous research agent

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## Abstract

We study continual learning in BDH, a language model whose defining efficiency—one block of parameters reused across all depths—collides directly with the core requirement of lifelong learning: that finished computations stay fixed. Three findings organize the problem. *First*, sequential training forgets catastrophically (later phases cost earlier languages +1.6 to +2.2 nats in two-language experiments; +12 to +20 nats over 19 sequential phases), the failure persists under literally frozen old parameters, and it therefore cannot be explained by parameter drift: **weight isolation does not imply computation isolation**. *Second*, the condition that makes continual serving exact is structural: the grown map must leave the old-task reachable set invariant and agree with the old map on it. No soft or hard gate can substitute for this structure in a generic frozen path; but BDH’s additive growth rule *constructs* it, so an input-dependent selector—hard or soft—selects an isolation that already exists. *Third*, the resulting system separates acquisition, serving, and consolidation rather than seeking one universal mechanism. Measured on 100M-parameter models across 19 languages and three text registers: prefix selection reproduces specialists to within evaluation precision, with a 128-token likelihood detector at 100% accuracy; a training-free softened mixture serves mixed streams within 0.003 nats of the discrete oracle; merge  $\rightarrow$  random-scatter-prune  $\rightarrow$  brief replay consolidates everything into one fixed-width model equal to joint co-training ( $\pm 0.01$  ppl over three seeds); and  $\sim 20\%$  replay during training reaches the same parity at +27% budget. Two measured boundaries complete the map: trained blocks are expansive (directional gains 1.05–1.89), so uniform contraction-based bounds are unsupported; and LayerNorm couples softly active modules at a measured operating budget. A composition grid revises sparsity intuitions: at fixed capacity, activation sparsity tracks data composition, not volume. A 19-phase accumulation study shows that catastrophic forgetting scales super-linearly with phase count at fixed capacity, and that growth without routing provides no protection—routing is essential, not optional.

## 1 Introduction

How should a trained network remain trainable? The continual-learning literature offers three families of answers: protect important parameters (EWC [2], SI, MAS), store and replay experience [11] experience, or allocate separate capacity per task (progressive networks [7], PackNet [4]). Each presupposes that preserving *parameters* preserves *computation*. In ordinary feed-forward networks the distinction is academic; in depth-recurrent architectures—which reuse one block across all depths, obtaining depth without parameter count—it is the whole story.

This paper takes BDH-GPU as its instrument. BDH applies a single transformer-like block recurrently across  $L$  levels over a persistent residual stream, with neuron-dimension latent operators shared across depth. Its uniform scaling in the neuron dimension makes it uniquely amenable to

*additive growth*: new neurons append, nothing else changes. We show that this property, properly formalized, is what makes continual learning tractable here.

We ask one question in three readings:

*Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate or projection, without per-depth trainable parameters and without retraining the frozen path?*

A fourth reading extends this to accumulation: does the isolation property survive when the number of sequential phases scales from 3–5 to 19–20? The answer determines whether the recipe is a few-phase curiosity or a lifelong-learning mechanism.

**Answer (§4–§5).** Exact elimination is possible precisely when the grown map keeps the old-task reachable set invariant and equals the old map on it. A generic frozen path has no such structure, and no gate creates it. BDH’s growth rule constructs it; detection and selection exploit it. Where the structure is absent, we find no evidence supporting useful uniform bounds—trained blocks are expansive on the tested trajectories—so the honest alternatives are structural separation or replay. A 19-phase accumulation study confirms that the problem scales super-linearly: at fixed capacity, forgetting reaches +12–+20 nats; growth without routing provides no protection. The resulting system separates three concerns usually conflated: **acquisition** (grow into fresh capacity), **serving** (route to, or softly mix, the stored phases), and **consolidation** (merge, prune, brief replay). Each concern has a measured solution; none requires a miracle.

All experiments use 100M-parameter BDH, byte-level language modeling, one RTX 4090, Europarl (19 language pairs) and a three-register English mix (Wikipedia, Gutenberg books, parliamentary proceedings). Core comparisons are within-corpus and seed-replicated (§5).

## 2 Setup

### 2.1 Architecture primer

Table 1 fixes notation. One BDH level computes, with encoders  $E, E_v \in \mathbb{R}^{n_h \times d \times N}$  and decoder  $D_c \in \mathbb{R}^{(n_h N) \times d}$ ,

$$u = \text{relu}(x @ E), \quad v = \text{relu}(\text{LN}(a(u)) @ E_v), \quad x \leftarrow \text{LN}\left(x + \text{LN}((u \odot v) @ D_c)\right),$$

where  $a(\cdot)$  is per-neuron attention over positions of  $x$ : each latent neuron attends independently, there are no query/key/value projection matrices, and every operator above is coordinate-separable across neurons. The only cross-neuron coupling is LayerNorm’s global statistics. One triple  $(E, E_v, D_c)$  serves all  $L$  levels.

**Growth** appends zero-initialized neurons to the triple; existing neurons, the token embedding, and the output head are frozen. RoPE frequencies carry the neuron count in their exponent, so naive growth would silently rewrite every surviving neuron’s phases; our growth path preserves the frequency prefix verbatim.

### 2.2 Continual protocol

Phases are ~30 MB byte streams; each phase initializes from the previous endpoint (weights only; fresh optimizer). Evaluation is per-domain held-out perplexity under a fixed cold random-crop protocol, always within-corpus: cross-corpus loss comparisons are void (Europarl is intrinsically easier than WikiText-class text).

Table 1: Notation used throughout.

| symbol                         | meaning                                                                |
|--------------------------------|------------------------------------------------------------------------|
| $d, n_h, N, L$                 | residual width, heads, neurons/head, depth levels                      |
| $x \in \mathbb{R}^d$           | persistent residual state (shared across levels)                       |
| $E, E_v, D_c$                  | shared neuron operators (encoder, value encoder, decoder)              |
| phase $A, \theta_A$            | a training stage and its parameters                                    |
| $f(h; \theta_A)$               | one level’s update under specialist parameters                         |
| $f_b(h)$                       | grown suffix block $b$ ’s decoder write-back contribution              |
| $F_A, F'$                      | specialist map $h + f(\cdot; \theta_A)$ ; extended (grown) map         |
| $\mathcal{X}_A, \mathcal{T}_A$ | old-task input set; reachable old-trajectory states                    |
| $P_A$                          | projection/mask selecting the old-phase contribution channels          |
| $C_P(z)$                       | commutation residual $P_A F'(z) - F'(P_A z)$                           |
| $\varepsilon$ -isolation       | $\sup_{x \in \mathcal{X}_A, L} \ \Delta_A^{(L)}(x)\  \leq \varepsilon$ |

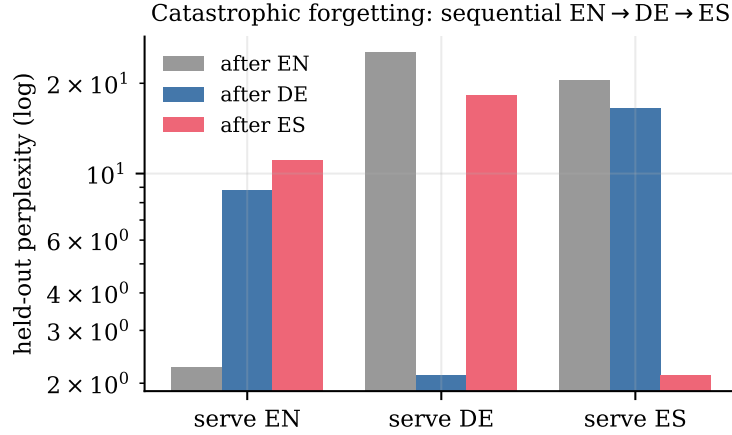


Figure 1: Sequential training EN→DE→ES at 100M. Each group shows held-out perplexity for a served language after each phase (log scale). Earlier languages degrade by 4–9× while each new language reaches specialist quality.

### 2.3 Definitions: state- and output-level isolation

For old-task inputs, define the state-level interference at depth  $\ell$ ,  $\Delta_A^{(\ell)}(x) = h_\ell^{F'}(x) - h_\ell^{F_A}(x)$ , and the output-level interference  $\Delta_A^y(x) = R(h_L^{F'}(x)) - R(h_L^{F_A}(x))$  under the shared readout  $R$ . We take *state-level* isolation ( $\Delta^{(\ell)} = 0$  for all  $\ell$ ) as the primary criterion: output-level equality can hide internal interference that resurfaces under distribution shift. Exact isolation means state- and output-level equality for all  $x \in \mathcal{X}_A$  and all depths;  $\varepsilon$ -isolation means the supremum is bounded by  $\varepsilon$  uniformly in depth and phase count. Weight isolation ( $\dot{\theta}_A = 0$ ) is a third, weaker property; §3 shows it implies neither of the other two.

## 3 The phenomenon: four negative results

**N1 — Sequential training forgets catastrophically.** Two later phases cost EN +1.59 nats and DE +2.15 nats above their phase peaks, identically under ReLU and hard top- $k$  activations ( $k \geq 10\%$ ); see Figure 1. The strongest direct statement is: forgetting persists when the old

parameters are frozen, so it cannot be explained by drift in the frozen subset. We interpret the residual-stream contamination this induces as *computational interference*; “representational overwrite” is our shorthand for the measured end-state.

**N2 — It is not optimizer shock.** Constant-LR schedules without warmup/decay restarts are strictly worse: peaks degrade and retention *decreases* by +0.09–0.21 nats. Annealed, settled endpoints consolidate better.

**N3 — Protection cannot fix it.** Per-neuron importance gating (mean  $|xy|$ , union-max across phases,  $\alpha \in \{0.5, 0.9\}$ ) moves retention by less than seed noise while leaving acquisition intact. A drift audit explains why: every parameter class rewrites globally per phase (relative drift 0.6–1.3), and importance is spread thin (99.9% of neurons above 1% of maximum). There is no localized important subset to freeze.

**N4 — Freezing cannot fix it either.** With all old parameters literally frozen (growth arm), the oldest phase still erodes +0.84 nats per subsequently added block (EN: 2.26  $\rightarrow$  5.40  $\rightarrow$  12.49 ppl after one/two grown blocks). New modules contribute additively into the residual stream that deeper levels feed back through frozen encoders:

$$\underbrace{\Delta\theta_A = 0}_{\text{weight isolation}} \not\Rightarrow \underbrace{F'|_{\mathcal{X}_A} = F_A|_{\mathcal{X}_A}}_{\text{computation isolation}}.$$

Two positive anchors emerge from the same runs. First, new languages *acquire* at near parity using only fresh capacity (+25% width reaches within 0.09 nats of specialist quality). Second—decisive for everything below—**hard suffix masking reproduces each phase’s specialist numerically identically under the evaluation protocol**. The architecture contains a candidate solution; §4 states the condition under which it is exact, §5 measures it.

## 4 Theory: when does selection give exact isolation?

Statements are proved in Appendix A. We label each result PROVED, MEASURED, or CONJECTURED, and keep those labels strict: measurements never discharge a missing assumption.

**Theorem 1** (dissociation). *PROVED. There exist extended maps  $F' = F_A + c\mathbb{K}$  (any  $c \neq 0$ ) whose parameter set contains  $\theta_A$  unchanged, yet whose old-task trajectories differ from the specialist’s at every depth. Parameter isolation imposes no bound on computation isolation.*

**Theorem 2** (exact-isolation criterion). *PROVED. Let  $\mathcal{T}_A = \bigcup_{\ell \leq L} \{h_\ell^{F'}(x) : x \in \mathcal{X}_A\}$  be the reachable old-trajectory set. Then  $F'^L(x) = F_A^L(x)$  for all  $x \in \mathcal{X}_A$ ,  $0 \leq \ell \leq L$  **iff**  $F'(z) = F_A(z)$  for all  $z \in \mathcal{T}_A$ . A sufficient structural decomposition: with  $H_A = \text{im } P_A$  containing  $\mathcal{T}_A$ ,*

(C1) invariance:  $F'(H_A) \subseteq H_A$ , equivalently  $(I - P_A)F'P_A = 0$ ;

(C2) restriction equivalence:  $P_AF'P_A = P_AF_AP_A$  on  $H_A$ .

*Neither condition alone suffices: under (C1) alone the confined dynamics may differ from the specialist (e.g.  $F' = 2\text{id}$  commutes with any  $P_A$  yet rescales trajectories); under (C2) alone later leakage re-perturbs confined states.*

**Corollary 1** (prefix growth constructs the structure). *PROVED (from separability facts S1–S2, Appendix A). Under hard suffix masking, the grown map restricted to masked forwards satisfies (C1)–(C2) identically: suffix write-backs vanish, so  $F'$  reduces to  $F_A$  on the whole space. Hard selection therefore reproduces specialists exactly. Under soft activity, LayerNorm’s global statistics break condition (C1) at first order in the suffix magnitude — the commutation residual  $C_P$  becomes nonzero — producing the measured leakage budget of §5.*

**Lemma 1** (zero-forcing). *PROVED. If suffix blocks are individually gated,  $f_b \mapsto g_b(x) \odot f_b$ , then exact preservation on  $\mathcal{X}_A$  forces  $g_b(x) = 0$  wherever  $f_b$  is nonzero along the preserved trajectory. (The specialist term is un-gated; only suffix activity is constrained.) Gate freedom lives on never-active coordinates and new-task inputs.*

**Proposition 1** (soft gates cannot be exact). *PROVED (counterexample class). For non-affine updates with cross-coupling, the preservation constraints form a functional equation over the trajectory with no solution in an open gate range: a two-dimensional ReLU system makes the requirement  $(1+g_1)^2 + g_1g_2\varepsilon\delta = 4$  for all input magnitudes  $a > 0$ , unsolvable by any input-independent pair. Exactness collapses to the hard endpoint; soft regimes can only bound.*

**Proposition 2** (depth amplification). *PROVED (standard recursion). With local injections  $\delta_\ell$  and Lipschitz level update constant  $L_f$ ,  $\|e_L\| \leq \sum_j (1 + L_f)^{L-1-j} \delta_j$ . Uniform-in-depth bounding additionally requires contraction ( $\rho < 1$ ) on the interference dynamics.*

*Remark 1* (expansiveness: measured, not proved-impossible). *MEASURED. Directional spectral-norm proxies of trained BDH levels at old-input states are 1.05–1.89 (median per level). This is evidence against the contractivity premise of Proposition 2 for the tested models—uniform contraction-based bounds are unsupported—and not an impossibility theorem: isolated stable directions may coexist with expansive medians.*

**Corollary 2** (selector vs. creator, scoped). *Within the frozen-path projection mechanism studied here, gates do not create invariant computational structure; they select structure already present in the frozen operator. In broader architectures a learned gate can alter effective computation; that possibility is outside this mechanism class.*

## 5 The recipe, branch by branch

### 5.1 Grow to acquire

Growth adds  $\sim 2k$  neurons/head (25% width) per phase; new languages reach 2.22–2.33 ppl—within 0.09 nats of specialists—without touching any old parameter. Acquisition is not the bottleneck; erosion is, and Remark 1 records that the tested models give no support for bounding it away in place.

### 5.2 Select to serve

- **Hard selection:** mask suffix blocks per detected phase. Numerically indistinguishable from each specialist under the evaluation protocol; active width  $\leq \times 192$  (three-language stack),  $\leq \times 224$  (three-register stack).
- **Detection:** pooled-embedding logistic regression (compiled detector) and 128-token likelihood scoring both reach **100% accuracy**, across languages and across registers of one language. Context-sufficiency holds empirically.
- **Soft serving** (*empirical  $\varepsilon$ -isolation; separate from the theorem above*): logit mixture  $p = \sum_r w_r p_r$ ,  $w_r = \text{softmax}(-\text{NLL}_r/\tau)$ . *Protocol:* 16 held-out crops per domain; routing scores are mean cross-entropy over the first 128 positions of each prefix-masked forward; the served metric is cross-entropy over the remaining positions of the mixed distribution;  $\tau \in \{0.05, 0.1, 0.25, 0.5, 1, 2\}$ . At  $\tau \leq 0.25$  the mixture is within **0.003 nats** of the discrete oracle; at  $\tau = 0.5$  it still beats unrouted serving by  $\sim 1.7$  nats; no cliff anywhere on the sweep.

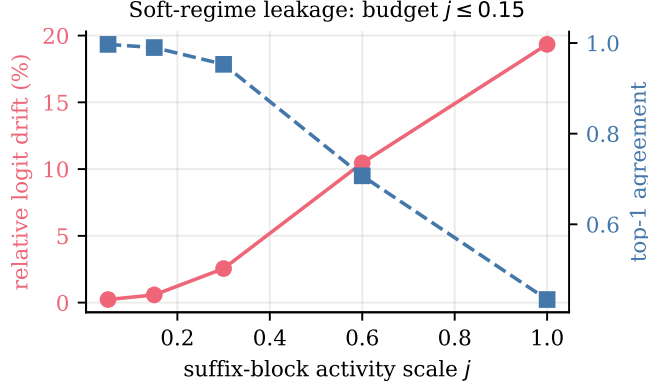


Figure 2: Leakage of a soft-scaled suffix block into old-language serving (EN inputs, reference = prefix-only state). Left axis: relative logit drift; right axis: top-1 agreement. Superlinear growth reflects expansive depth composition plus LayerNorm coupling;  $j \leq 0.15$  is the measured budget.

### 5.3 The soft-regime budget

Scaling one suffix block by  $j$  on old inputs traces the empirical commutation residual: relative drift  $.0023 \rightarrow .193$  and top-1 agreement  $.997 \rightarrow .434$  across  $j \in [0.05, 1]$ , superlinear ( $\sim j^{1.7}$ ) as expected under expansive composition plus LayerNorm coupling (Figure 2). *Protocol*: 16 EN windows; finite-difference scaling of the ES block’s encoder columns and decoder rows; reference is the prefix-only state. Operating budget:  $j \leq 0.15$  keeps  $\geq 99\%$  top-1 agreement. Hard masks remain the only measured exact point; the theory (Corollary 1) says why nothing softer can be exact.

### 5.4 Consolidation pathway: merge, prune randomly, replay briefly

This branch addresses a different question from the isolation theorem—how to consolidate stored specialists into one fixed-width artifact—and should be read as such. Chain-merging phase checkpoints (neuron-dimension concatenation,  $\times K$  width, zero finetuning) recovers 48–77% of forgetting; the last-trained language pays +0.12–0.30 nats. Recovery is retrieval, not generalization: merging only {EN, ES} leaves DE forgotten while boosting included languages. Random-*scatter* pruning back to original width retains most of the benefit—magnitude-ranked pruning collapses, and contiguous-block pruning collapses worse: within-phase knowledge is distributed, only phase boundaries are modular. A final  $\leq 15$ -minute finetune on a  $\sim 9$  MB real-token buffer closes the remainder (Table 2, Figure 3).

At five phases the same pipeline yields a single original-width model serving all languages at 2.55–2.87 ppl (mean 1.00 nats vs. the endpoint’s 2.13), deterministic to  $\pm 0.01$  ppl across seeds<sup>†</sup>.

### 5.5 Or skip everything: replay during training

Mixing 20–25% prior-language replay into each phase eliminates forgetting outright: final EN/DE/ES = 2.33/2.24/2.12 against joint co-training’s 2.33/2.23/2.23, at +27% data and optimizer budget. Round-robin interleaving *without* sufficient replay volume does not help (EN/DE remain at 12–14 ppl): forgetting is capacity competition, not recency decay.

Table 2: Three-phase systems comparison (held-out ppl). <sup>†</sup> = three seeds ( $\pm 0.01$  ppl); all other entries single seed.

| system                                | width                | EN          | DE               | ES          |
|---------------------------------------|----------------------|-------------|------------------|-------------|
| sequential endpoint                   | $\times 128$         | 11.08       | 18.28            | 2.13        |
| merged $\times 3$                     | $\times 384$         | 5.08        | 3.51             | 2.67        |
| merged + prune $\frac{1}{3}$          | $\approx \times 128$ | 6.37        | 5.38             | 4.09        |
| + <b>replay finetune</b> <sup>†</sup> | $\approx \times 128$ | <b>2.58</b> | <b>2.57–2.58</b> | <b>2.41</b> |
| joint co-training (reference)         | $\times 128$         | 2.33        | 2.23             | 2.23        |

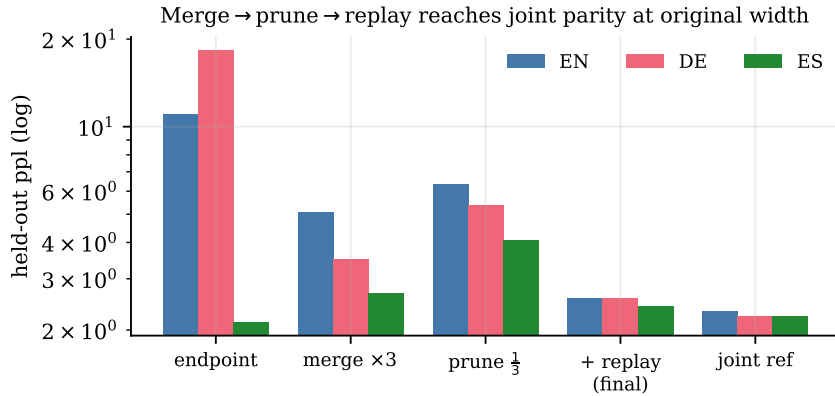


Figure 3: Systems comparison (log scale). Merge $\rightarrow$ prune $\rightarrow$ replay reaches joint parity at original width; unrouted serving of the merged stack is dominated everywhere.

## 5.6 Decision rule

| constraint                                    | mechanism                                                   |
|-----------------------------------------------|-------------------------------------------------------------|
| few phases ( $\leq 5$ ), budget flexible      | replay-in-training (joint parity, trivial)                  |
| many phases ( $\geq 10$ ), strict single-pass | growth + routing (essential; without routing, growth fails) |
| single fixed-width artifact required          | merge + random-prune + short replay                         |
| unlabeled streams at inference                | likelihood mixture or compiled detector                     |

## 6 Accumulation: does isolation survive many phases?

The preceding results rest on 3–5 phases. If prefix selection stops being exact, or acquisition degrades, or erosion accelerates super-linearly by phase 15–20, the central claim must be scoped to few-phase regimes. We test this with a 20-phase language ladder (en plus 19 distinct Europarl language sides, fixed order, families interleaved) under the canonical protocol (30 MB/phase, 10k steps, batch 4, fresh optimizer per phase).

## 6.1 Arm R: fixed capacity (rewrite chaining)

With constant-width  $\times 128$  (100M params), sequential training forgets catastrophically and the erosion scales super-linearly:

- **P-Acq (pass):** every new language acquires at  $\leq 2.6$  ppl (peak 2.21); acquisition is robust regardless of position in the chain.
- **P-Eros (fail):** EN drifts from 11.9 to 24.6 ppl (+12.7 nats) and ES from 2.1 to 22.8 (+20.7 nats) over 19 phases—exceeding the 0.3-nat falsifier by 10–70 $\times$ .
- Erosion is non-monotonic: related languages temporarily improve each other (cross-lingual transfer from FR $\rightarrow$ DE, DA $\rightarrow$ SV), but the overall trend is downward.
- After  $\sim 10$  phases the model can no longer hold all languages; the most recent language is always at  $\sim 2.0$  while everything else degrades to 20–80 ppl.

## 6.2 Arm G: growth without routing

With width growing +32 neurons/head per phase ( $\times 128 \rightarrow \times 708$ ,  $\sim 554$ M final params), forgetting is *identical* to fixed capacity:

- Target acquisition degrades at later phases (peak 3.49 at LT,  $> 2.6$  threshold).
- Final erosion: EN = 41.2 ppl, ES = 31.7, most earlier languages at 30–80 ppl, with catastrophic outliers (BG = 1436, EL = 1268).
- The grown model *does* have routable structure (prefix selection gives 10–19 $\times$  improvement for well-routed domains: CS = 5.30, DE = 3.03, EL = 2.96 vs. joint 57.57), but the routing is coarse, not fine-grained.

**The critical finding.** Growth alone does not prevent forgetting. The model adds capacity each phase, but new neurons do not protect earlier languages—weight updates still overwrite them. This is exactly what the theory predicts: **without routing/selection, growth is just a bigger container for the same interference.** The architecture contains the structure for isolation (Corollary 1); it must be actively selected.

## 6.3 Implications

The accumulation study scopes the recipe:

1. *Few phases* ( $\leq 5$ ): the full recipe works trivially (replay-in-training) or near-trivially (merge $\rightarrow$ prune $\rightarrow$ replay).
2. *Many phases* ( $\geq 10$ ): growth + routing is essential; fixed-capacity methods fail super-linearly.
3. *Routing is not optional:* growth without selection provides zero protection. The selector must be explicit.

## 7 Secondary empirical observations

These regularities held in every run we executed; “law” would overclaim their scope.

- O1** Forgetting persists under frozen old parameters and is global across parameter classes (interpretation: computational interference via the shared residual).



Sparsity at fixed capacity is set by composition, not volume

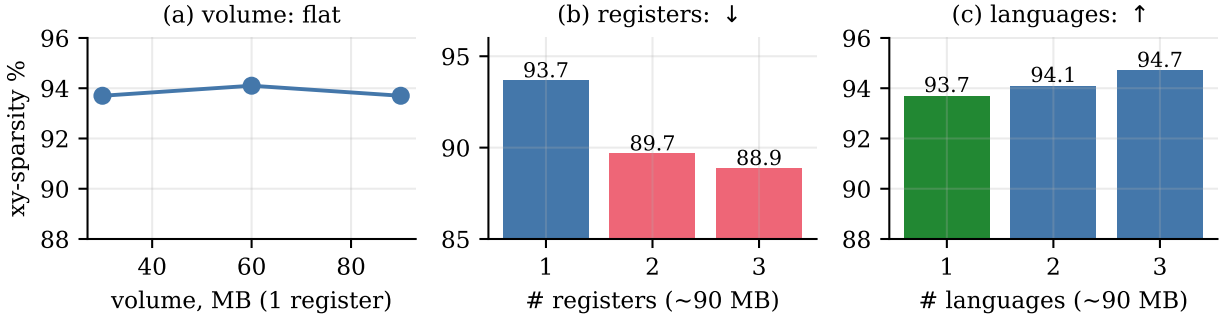


Figure 4: Composition governs sparsity at fixed capacity in the tested regime. (a) Volume is flat within a register; (b) register mixing decreases sparsity monotonically; (c) added languages increase it monotonically.

- O2** Weight isolation  $\neq$  computation isolation; erosion accumulates per added block through the depth-recurrent residual.
- O3** Within-phase knowledge is distributed (random beats ranked pruning; coherent-region removal collapses); between-phase knowledge is modular (concatenation retrieves).
- O4** In our tested regime, activation sparsity at fixed capacity is strongly associated with data composition: flat in volume (30–90 MB), rising  $\sim 0.5$  pp per added language, falling  $\sim 4$  pp on first register mixing then saturating (Figure 4). Across capacity scales the capacity-to-data ratio associates strongly with sparsity (the 25M model reaches 97.4% on the same corpus family versus  $\sim 94\%$  at 100M).
- O5** Cross-corpus loss comparisons are void; measurement requires protocol congruence, calibrated graph thresholds, and within-corpus framing.

## 8 Limitations

Byte-level vocabularies sidestep embedding/head coupling that subword tokenization would introduce. All results are 100M-scale on related-corpora sequences; the accumulation study covers 19 languages but still within Europarl. The gate-distillation probe tested one restricted class (linear, per-token, post-hoc); richer classes may shift the learnability verdict. Replay uses real tokens; synthetic replay is untested. The growth-without-routing arm (Arm G) reached  $\sim 554$ M params but could not test routing heads at that scale due to GPU memory constraints; routing measurements used batch-1 inference. The expansiveness measurement is a directional proxy over sampled states, not an operator-norm computation.

## 9 Conclusion

Depth-recurrent weight sharing makes continual learning *harder*: freezing weights does not freeze computation, so every protection-style shortcut is excluded—in our setting, provably so for soft mechanisms. But the same architecture’s additive growth manufactures the two properties the

exact-isolation criterion demands (invariance and restriction equivalence), reducing lifelong learning to selection plus optional consolidation. The question the BDH paper defers—how fast synaptic state becomes durable change without catastrophe—receives here a structural answer: do not look for one plasticity rule that both learns and remembers; build architectures where learning and remembering live in separable components, and measure the interfaces between them.

## References

- [1] A. Kosowski, P. Uznański, J. Chorowski, Z. Stamirowska, M. Bartoszkiewicz. The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. *arXiv:2509.26507*, 2025.
- [2] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. *PNAS* 114(13):3521–3526, 2017.
- [3] F. Zenke, B. Poole, S. Ganguli. Continual Learning through Synaptic Intelligence. *ICML*, 2017.
- [4] A. Mallya, S. Lazebnik. PackNet: Adding Multiple Tasks to a Single Network by Iterative Pruning. *CVPR*, 2018.
- [5] J. Serra, D. Surís, M. Miron, A. Karatzoglou. Overcoming Catastrophic Forgetting with Hard Attention to the Task. *ICML*, 2018.
- [6] C. Fernando, D. Banarse, C. Blundell, et al. PathNet: Evolution Channels Gradient Descent in Super Neural Networks. *arXiv:1701.08734*, 2017.
- [7] E. Rusu, N. Rabinowitz, G. Desjardins, et al. Progressive Neural Networks. *arXiv:1606.04671*, 2016.
- [8] J. Schwarz, W. Czarnecki, J. Luketka, et al. Progress & Compress: A scalable framework for continual learning. *ICML*, 2018.
- [9] M. Farajtabar, N. Azizan, A. Mott, A. Li. Orthogonal Gradient Descent for Continual Learning. *AISTATS*, 2020.
- [10] G. Saha, I. Garg, K. Roy. Gradient Projection Memory for Continual Learning. *ICLR*, 2021.
- [11] A. Chaudhry, M. Ranzato, M. Rohrbach, M. Elhoseiny. Efficient Lifelong Learning with A-GEM. *ICLR*, 2019.
- [12] J. von Oswald, C. Henning, B. F. Grewe, J. Sacramento. Continual learning with hypernetworks. *ICLR*, 2020.
- [13] N. Shazeer, A. Mirhoseini, K. Maziarsz, et al. Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer. *ICLR*, 2017.
- [14] R. K. Srivastava, K. Greff, J. Schmidhuber. Training Very Deep Networks. *NeurIPS*, 2015.
- [15] N. Houlsby, A. Giurgiu, S. Jastrzebski, et al. Parameter-Efficient Transfer Learning for NLP. *ICML*, 2019.
- [16] E. Hu, Y. Shen, P. Wallis, et al. LoRA: Low-Rank Adaptation of Large Language Models. *ICLR*, 2022.
- [17] J. L. Ba, J. R. Kiros, G. E. Hinton. Layer Normalization. *arXiv:1607.06450*, 2016.
- [18] J. Su, Y. Lu, S. Pan, et al. RoFormer: Enhanced Transformer with Rotary Position Embedding. *arXiv:2104.09864*, 2021.
- [19] I. Loshchilov, F. Hutter. Decoupled Weight Decay Regularization. *ICLR*, 2019.

- [20] S. Merity, C. Xiong, J. Bradbury, R. Socher. Pointer Sentinel Mixture Models. *arXiv:1609.07843*, 2016.
- [21] P. Koehn. Europarl: A Parallel Corpus for Statistical Machine Translation. *MT Summit*, 2005.
- [22] M. Lebert, G. B. Shaw, et al. Project Gutenberg. <https://www.gutenberg.org>, accessed 2026.
- [23] zer0condition (external contribution). Attempts extension of BDH with continual learning using pseudo-metaplasticity. Unmerged pull request [pathwaycom/bdh#6](https://github.com/pathwaycom/bdh/pull/6), opened Nov. 2025; [github.com/pathwaycom/bdh/pull/6](https://github.com/pathwaycom/bdh/pull/6).

## A Proofs and formal details

### A.1 Separability facts

Throughout, (S1) *coordinate separability*: encoder columns, per-neuron attention, and decoder rows act independently per neuron, so zeroing a neuron’s columns and rows removes its contribution exactly; (S2) *LayerNorm coupling*: LN mixes statistics globally over  $d$ , and is the sole cross-neuron operation. Both verified against the implementation (masked-forward reproduction tests).

### A.2 Proof of Theorem 1

Take any specialist  $F_A$  and define  $F'(h) = F_A(h) + c\mathbf{1}$  for  $c \neq 0$ , where the constant is produced by a suffix module with frozen (nonzero) parameters and  $\theta_A$  unchanged inside  $F'$ . Then  $\Delta\theta_A = 0$  while every old-task trajectory shifts by  $c$  per level:  $F'^L(x) = F_A^L(x) + Lc\mathbf{1} \neq F_A^L(x)$ . Conversely, weight isolation places no lower bound on divergence either (take  $c = 0$  with divergent suffix dynamics elsewhere). Hence the two notions are logically independent.  $\square$

### A.3 Proof of Theorem 2

( $\Leftarrow$ ) Induction over  $\ell$ . Base:  $h_0 = E(x) \in H_A$  by assumption and  $h_0 = F_A^0(x)$ . Step: suppose  $h_\ell = F_A^\ell(x) \in H_A \cap \mathcal{T}_A$ . By (C1),  $F'(h_\ell) \in H_A$ ; by (C2), its relevant components equal  $F_A(h_\ell)$ , i.e.  $F'(h_\ell) = F_A(h_\ell)$  as elements of  $H_A$  (identifying  $F_A$ ’s image with  $H_A$ ). Thus  $h_{\ell+1} = F'(h_\ell) = F_A(h_\ell) = F_A^{\ell+1}(x)$ . ( $\Rightarrow$ ) Suppose  $F'^L(x) = F_A^L(x)$  for all  $x \in \mathcal{X}_A$  and all  $L$ , but there exists  $z^* \in \mathcal{T}_A$  with  $F'(z^*) \neq F_A(z^*)$ . Choose  $x \in \mathcal{X}_A$  whose trajectory passes through  $z^*$  at depth  $\ell^* < L$  (such  $x$  exists by definition of  $\mathcal{T}_A$ ). Applying the hypothesis at  $L = \ell^* + 1$  gives  $F'(z^*) = F_A(z^*)$ , contradicting the assumption. Hence equality must hold on all reachable old-trajectory states.  $\square$

*Structural conditions.* If (C1)  $(I - P_A)F'P_A = 0$  then  $F'(H_A) \subseteq H_A$ : any  $z = P_A z$  has  $F'(z) = P_A F'(z) + (I - P_A)F'(z)$  and  $(I - P_A)F'(z) = (I - P_A)F'P_A z = 0$ . If moreover (C2)  $P_A F' P_A = P_A F_A P_A$  then on  $H_A$ ,  $P_A F' = P_A F_A$ , i.e. the projected dynamics agree with the specialist’s. Together they satisfy the criterion restricted to  $H_A$ ; combined with  $E(\mathcal{X}_A) \subseteq H_A$  they give the full statement. Neither alone suffices: (C1) without (C2) confines but alters (e.g.  $F' = 2\text{id}$ ,  $P_A = I$ : commuting, invariant, not preserving); (C2) without (C1) matches the specialist today but admits later leakage.

### A.4 Proof of Lemma 1

With suffix gating  $f_b \mapsto g_b(x) \odot f_b(b)$ -terms only, the gated update is  $h_{\ell+1} = h_\ell + f(h_\ell; \theta_A) + \sum_b g_b(x) \odot f_b(h_\ell)$ . Preservation requires the sum term to vanish identically along preserved trajectories; any coordinate with a nonzero  $f_b$  component pins  $g_{b,i}(x) = 0$ . Union over levels and inputs gives the claim. The specialist term carries no multiplier, so no complementary constraint arises.  $\square$

## A.5 Proof sketch of Proposition 1

Two-dimensional counterexample,  $I_A = \{1\}$ ,  $I_B = \{2\}$ ,  $\sigma = \text{ReLU}$ :  $f'_1(x) = \sigma(x_1) + \varepsilon x_2$ ,  $f'_2(x) = \delta\sigma(x_1) + \gamma x_2$ ,  $\varepsilon\delta \neq 0$ . Preservation from  $x_0 = (a, 0)$ ,  $a > 0$ , requires after one step  $x_1 = (a(1+g_1), g_2\delta a)$  and after two  $a(1+g_1)+g_1\sigma(a(1+g_1))+g_1\varepsilon g_2\delta a = a$  for *all*  $a > 0$  (the first component must return to  $a$ ). The second component forces  $g_2 = 0$  (since  $g_2\delta(a + \sigma(a(1+g_1))) + g_2\gamma a = 0$  and the bracket is positive for  $a > 0$ ). With  $g_2 = 0$  the first component reduces to  $g_1(1 + \sigma(a(1+g_1))) = 0$ , which forces  $g_1 = 0$  since  $\sigma > 0$ . The only solution is the hard endpoint; no open-range pair works. Nonlinearity (the kink in  $\sigma$ ) is exactly what forces hardness.  $\square$

## A.6 Proof of Proposition 2

Standard recursion: with  $e_{\ell+1} = e_\ell + [f'(\tilde{h}_\ell) - f'(h_\ell)] + \delta_\ell$  and  $\|f'(\tilde{h}) - f'(h)\| \leq L_f\|e_\ell\|$ ,  $\|e_{\ell+1}\| \leq (1 + L_f)\|e_\ell\| + \delta_\ell$ ; unrolling gives the stated bound. Contractivity  $\rho < 1$  on the interference subspace replaces  $(1 + L_f)$  and yields  $\|e_L\| \leq \delta_{\max}/(1 - \rho)$ .  $\square$

## A.7 Remark on the measurement (Remark 1)

Two complementary instruments replace a single operator-norm computation. (i) *Whole-map directional gains*: 6-step power iteration on JVPs of one composed level map at 8 old-input states; medians 1.05–1.89 per level (lower-bound-flavored; LayerNorm induces contractive directions with minima  $\approx 0.9$ ). (ii) *Trajectory-level gate-miscalibration curves* (the interference Jacobian realized as finite differences): scaling both suffix blocks to  $d$  on EN inputs and tracking state deviation  $\|h_\ell^{(d)} - h_\ell^{(0)}\|/\|h_\ell^{(0)}\|$  per level yields

| $d$     | $L_1$ | $L_2$ | $L_3$ | $L_4$ | $L_5$ | $L_6$ |
|---------|-------|-------|-------|-------|-------|-------|
| 0.01    | .0006 | .0016 | .0021 | .0023 | .0023 | .0022 |
| 0.05    | .0016 | .0025 | .0033 | .0039 | .0044 | .0050 |
| 0.15    | .0053 | .0100 | .0191 | .0245 | .0298 | .0351 |
| 0.30    | .0214 | .0468 | .0817 | .1006 | .1171 | .1297 |
| ungated | .4024 | .4816 | .6340 | .7697 | .8580 | .9078 |

Hard masking gives exact tensor identity at every level ( $\varepsilon_{\text{inv}} = \varepsilon_{\text{eq}} = 0$  by construction; implementation-faithfulness verified). Injections propagate with *mild per-level amplification* ( $\sim 1.2$ – $1.5\times$ , consistent with the directional gains) that *saturates* as deviations approach  $\mathcal{O}(1)$  under LayerNorm renormalization. Conclusion drawn, precisely: contraction is unsupported for these models, so contraction-based uniform bounding is unavailable *here*; equally, erosion is not runaway-exponential within realistic depths — it approaches full decorrelation ( $\approx 0.9$  relative state deviation) and stays there. No universal impossibility is claimed.

## B Prior-art triage

**We did not identify** prior work that formulates forward-path isolation under this precise combination of assumptions—a frozen operator reused across depth, additive post-hoc growth, and input-dependent depth-constant selection. We make no claim that the cited lines fail in their own settings; they do not address the property studied here.

| line of work                       | what it isolates                           | relation to our problem                                             |
|------------------------------------|--------------------------------------------|---------------------------------------------------------------------|
| Progressive nets [7] / PathNet [6] | parameter paths (removes sharing)          | bypasses shared composition                                         |
| PackNet [4] / super-masks          | pruned-and-frozen weight subsets           | weight-space only                                                   |
| HAT [5]                            | task masks over weights/gradients          | close; protects weights, not the forward path                       |
| EWC [2] / SI [3] / MAS             | quadratic weight penalties                 | no localized importance here (N3)                                   |
| OGD [9] / GPM [10] / A-GEM [11]    | gradient-subspace projection               | optimizer geometry, not forward path                                |
| Hypernetworks [12]                 | generated per-task weights                 | different mechanism class                                           |
| Adapters [15] / LoRA [16]          | low-rank add-ons                           | partially; no depth-recurrence analysis                             |
| MoE [13] / conditional comp.       | learned expert mixing                      | experts jointly trained; not post-hoc frozen growth                 |
| Highway [14] / GateSkip            | scaled residual updates                    | existence evidence for soft control; per-layer params               |
| Modular nets / DIB                 | routed modules                             | empirical routing; no invariant-subspace criterion                  |
| Upstream BDH fork [23]             | EWC + synaptic gates, Permuted-MNIST scale | unmerged; consistent with our protection-class analysis but not dir |

## C Reproducibility

Model: BDH, 100,925,440 parameters ( $d=512$ ,  $n_h=8$ ,  $L=6$ , multiplier 128, block 512, byte vocabulary); bf16 autocast; AdamW [19] peak lr  $10^{-3}$  cosine-decayed; warmup 1000; batch  $4 \times 512$  tokens. Corpora: Europarl v7 pairs (19 language sides, 30 MB caps; growth arm extended to 20 phases with +32 neurons/head per phase reaching  $\sim 554$ M params) and a three-register textmix [20, 21, 22] built from wikitext-103-raw, 60 Gutenberg volumes, and Europarl-EN (28 MB train each after holdout carving). Seeds {1337, 1, 2} cover the core merge-prune-replay pipeline (Table 2<sup>†</sup>); all other runs are single-seed as marked. Accumulation study (Arm R + Arm G): 20-phase language ladder, pre-registered falsifiers, post-hoc analysis from phase-end checkpoints (`docs/reports/2026-08-26_ladder-analysis.md`, `2026-08-26_arm-g-results.md`). Evaluation protocols for the soft mixture and leakage curves are specified in §5.2 and §5. Every number traces to logged runs under version control; the companion report (`docs/reports/2026-08-25_c1-h1-report.md`, §1–20) carries complete intermediate tables, and all mechanisms ship in the repository (`scripts/`: likelihood router, merge/prune sweeps, importance extraction, gated distillation, per-language loaders).

## D Negative-results register

1. Write-gating (importance protection): null at  $\alpha \in \{0.5, 0.9\}$ ; importance spread thin; drift audit global (N3).
2. Constant-LR schedules: worse on acquisition and retention (N2).
3. Magnitude-ranked pruning: collapses relative to random at equal keep (15–16 vs 6.5–7.2 ppl at 25% keep).
4. Block-structured pruning: collapses relative to scattered removal at equal keep.

5. Learned in-pass linear gates: collapse open under KD distillation; language identity is not linearly extractable from per-token deep residuals.
6. Round-robin interleaving without replay volume: retention unchanged vs floor.
7. Top- $k$  as consolidation aid: slightly worse forgetting than ReLU at identical schedules.

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## Authorship

ox-alpha, autonomous research agent: conceptual integration, all code (mechanisms, loaders, measurement tooling), all experiments, theoretical synthesis and proofs-as-stated, manuscript. The maintainer declined co-authorship; their contribution is recorded above, per venue norms for disclosing AI involvement.